Forces on two objects

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Identify an equal-and-opposite interaction pair on different objects.

SUCCESS CRITERIA - I can...

1. Name both objects in an interaction pair.
2. Place the two equal-and-opposite forces on different objects.
3. Explain why equal forces can produce unequal accelerations.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Warm-up: bring back an earlier idea

A **0.4 kg** system accelerates at **0.6 m/s²**.
1. **Predict acceleration** for **0.8 kg** at the same net force.
2. **Explain the effect of greater** resistance.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $F_{A \text{ on } B} = -F_{B \text{ on } A}$.
The two forces act on different objects at the same instant. Their accelerations can differ.

Worked reasoning: Two skaters push apart. Each feels **20 N** from the other. If their masses are **40** and **80 kg**, accelerations are **0.5** and **0.25 m/s²** in opposite directions.

Your turn: 1. **Draw separate force diagrams** for the horizontal interaction.
2. **Explain why equal forces** do not imply equal accelerations.



Skater A: 40 kg



Skater B: 80 kg

Marzano check - circle: ① ② ③

03 Equal forces, different results

On a low-friction track, a **1 kg** cart and a **4 kg** cart push apart. Each interaction force is **12 N**. **Draw the force** on each cart, calculate both acceleration magnitudes, and **explain why the forces match** but the accelerations do not.



Cart A: 1 kg



Cart B: 4 kg

Marzano check - circle: ① ② ③

04 Name actual partners

For a book on a table, name the partner of Earth-on-book and the partner of table-on-book. **Explain why weight** and normal are not a third-law pair.

Marzano check - circle: ① ② ③

Forces on two objects

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Continue down each column.

05 Horse plus cart

A horse pulls a cart forward while the cart pulls backward on the horse with equal force.
1. **Draw a boundary around the** two-object system.
2. **Add the ground's force** on the horse and any resistance.
3. **Explain how the combined system** can accelerate forward.



Horse



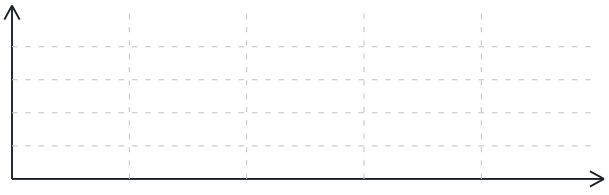
Cart

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

East is positive. Start at **7 m**, go to **-5 m**, then finish at **4 m**. **Sketch the route**; **calculate distance** and displacement. Distinguish both from final position. **Label the diagram** with the given positions and any prediction.

Route



Position (m)

Marzano check - circle: 1 2 3

07 Exit check

A spacecraft pushes an asteroid that has much greater mass.
1. **Draw the interaction force** on each object.
2. **Compare their forces** and the acceleration caused by this interaction.
3. **State what you assume about** other forces.



Spacecraft

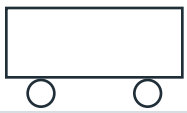


Asteroid

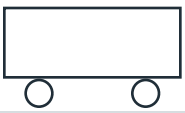
Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A **2 kg** cart pushes a **3 kg** cart along a frictionless track. An external **10 N** force pushes the **2 kg** cart to the right. **Draw separate force diagrams**, find their shared acceleration, and **calculate the contact force**.



2 kg; applied 10 N right



3 kg

Marzano check - circle: 1 2 3